

E084

Advanced C#

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

Method © 2026 Erik Daniel Lindquist — Course Mastery Coach, armored drill. Items drafted with AI assistance under his direction.

How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A pricing service at an airline with 700 aircraft calls three independent APIs that take 300 ms, 500 ms, and 200 ms. The developer, who joined from a game studio, starts all three tasks and then awaits Task.WhenAll. About how long does the combined wait take?

- A. About 200 ms
- B. About 333 ms
- C. About 1,000 ms
- D. About 500 ms

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

Code awaits three independent calls one after another and takes about 1,000 ms. Which change brings it close to the slowest call's time?

- A. Mark the method async void.
- B. Use .Result on each call instead of await.
- C. Wrap each call in Task.Run but await each before starting the next.
- D. Start all three tasks, then await Task.WhenAll on them.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A logging helper at a busy e-commerce site, written the week before a holiday sale, is declared async void SaveAsync(). When the database is down, the app crashes even though the caller wraps SaveAsync() in try/catch. What is the best fix?

- A. Add a second try/catch around the call.
- B. Add ConfigureAwait(false) to the database call.
- C. Change it to async Task and await it inside the try/catch.
- D. Never use async void anywhere, including event handlers.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

An import tool for a school district with 60 schools opens a FileStream, parses a large CSV, and sometimes throws on a bad row, leaving the file locked. The CSVs arrive every night at 2 a.m. Which change guarantees the file is released?

- A. Call stream.Dispose() on the last line of the method.
- B. Open the stream in a using statement or declaration.
- C. Set the stream variable to null after parsing.
- D. Let the garbage collector close it.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A reporting class at an insurance firm, reviewed by two senior developers, runs: `var nums = new List<int> { 3, 6, 8 }; var big = nums.Where(n => n > 5); nums.Add(9); Console.WriteLine(big.Count());` What prints?

- A. 4
- B. 3
- C. 2
- D. An exception, because the collection was modified.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

A developer wants a LINQ result that will not change if the source list changes later. What should they do when defining it?

- A. Store the query in a var and reuse it.
- B. Make the lambda static.
- C. Call ToList() on the query.
- D. Call AsEnumerable() on the query.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A greeting service for a gym chain with 30 locations runs `var name = customer?.Profile?.Name ?? "Guest";` for a customer whose Profile is null. The gym opens at 5 a.m. What is name?

- A. A `NullReferenceException` is thrown.
- B. "Guest"
- C. null
- D. An empty string

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A product-search endpoint for a hardware retailer, built during a two-week sprint, adds the user's search text directly into a SQL string. Only logged-in users can search. What is the best defensive fix?

- A. Use parameterized queries and validate input against an allow-list.
- B. Nothing, because input from logged-in users is always safe.
- C. Manually escape single quotes in the text.
- D. Block a short list of known bad words.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

An exception is thrown inside a method that checks whether a user may view a record. Which failure behavior is safe?

- A. Retry the check forever until it succeeds.
- B. Return the full stack trace to the user to help debugging.
- C. Deny access, log the details internally, and return a generic error.
- D. Allow access so the user is not blocked by a bug.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A SummaryService class at a legal-tech startup with an award-winning logo creates new OpenAiClient() inside its constructor. Unit tests are slow and cost money on every run. The startup has 14 employees. What design change helps most?

- A. Depend on an IChatClient interface passed in through the constructor.
- B. Have SummaryService inherit from OpenAiClient.
- C. Make the client a static singleton.
- D. Avoid interfaces; they always slow code down.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A checkout bug at a pet-supply store with a popular blog applied a discount twice. A developer fixed it on a Friday afternoon. Code coverage is already 85%. What should be added so the bug cannot quietly return?

- A. A one-time manual check of the checkout page.
- B. Only end-to-end browser tests.
- C. Nothing; 85% coverage already guarantees no regressions.
- D. An automated regression test that reproduces the double discount, run in CI.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

In a healthy automated test suite, which type of test should be the most numerous?

- A. Unit tests
- B. Manual exploratory tests
- C. End-to-end UI tests
- D. Equal numbers of every type, always

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer D Concept 1

A pricing service ~~at an airline with 700 aircraft~~ calls three independent APIs that take 300 ms, 500 ms, and 200 ms. The developer, ~~who joined from a game studio~~, starts all three tasks and then awaits Task.WhenAll. About how long does the combined wait take?

Target word: total time with WhenAll **Decides:** Concurrent tasks finish when the slowest does.

A	Inverted operation	Uses the fastest task; the others are still running.
B	Confusion set	The average duration does not decide completion time.
C	Adjacent category	That is sequential awaiting: 300 + 500 + 200.
D	Correct	The slowest task (500 ms) sets the total.

2. Answer D Concept 1 · transfer

Code awaits three independent calls one after another and takes about 1,000 ms. Which change brings it close to the slowest call's time?

Target word: reverse: pick the change **Decides:** Tasks must be started before any is awaited.

A	Confusion set	Changes error handling, not concurrency.
B	Adjacent category	Blocks the thread and still runs them one at a time.
C	Dropped element	Still sequential; nothing overlaps.
D	Correct	The calls overlap, so total $\approx$ the slowest one.

3. Answer C Concept 2

A logging helper ~~at a busy e-commerce site, written the week before a holiday sale~~, is declared async void SaveAsync(). When the database is down, the app crashes even though the caller wraps SaveAsync() in try/catch. What is the best fix?

Target word: why the catch fails and the fix **Decides:** Exceptions from async void cannot be awaited or caught by the caller.

A	Confusion set	Any number of caller catches miss async void exceptions.
B	Adjacent category	Changes the resume context, not exception flow.
C	Correct	A returned Task carries the exception to the awaiting caller.
D	Absolute language	Event handlers are the accepted exception.

4. Answer B Concept 3

An import tool ~~for a school district with 60 schools~~ opens a FileStream, parses a large CSV, and sometimes throws on a bad row, leaving the file locked. ~~The CSVs arrive every night at 2 a.m.~~ Which change guarantees the file is released?

Target word: guaranteed cleanup **Decides:** Cleanup must run even when an exception is thrown.

A	Dropped element	Skipped when an exception is thrown earlier.
B	Correct	Dispose runs automatically when the scope exits, error or not.
C	Confusion set	Clearing a reference does not release the handle.
D	Adjacent category	Collection timing is not deterministic.

5. Answer B Concept 4

A reporting class at an insurance firm, reviewed by two senior developers, runs: `var nums = new List<int> { 3, 6, 8 }; var big = nums.Where(n => n > 5); nums.Add(9); Console.WriteLine(big.Count());` What prints?

Target word: count after the source changes **Decides:** Where is deferred; it runs at Count(), after 9 is added.

A	Wrong stem number	The list's total count, ignoring the filter.
B	Correct	At enumeration the list is {3, 6, 8, 9}; 6, 8, and 9 exceed 5.
C	Confusion set	Assumes the query ran when it was defined.
D	Adjacent category	That error occurs only when modifying during enumeration.

6. Answer C Concept 4 · transfer

A developer wants a LINQ result that will not change if the source list changes later. What should they do when defining it?

Target word: force immediate execution **Decides:** A snapshot requires materializing the results.

A	Dropped element	The var holds the query, not its results.
B	Adjacent category	Affects captured variables, not when the query runs.
C	Correct	Runs the query now and stores the results.
D	Confusion set	Still deferred.

7. Answer B Concept 5

A greeting service for a gym chain with 30 locations runs `var name = customer?.Profile?.Name ?? "Guest";` for a customer whose Profile is null. The gym opens at 5 a.m. What is name?

Target word: value of a null-safe chain **Decides:** ?. yields null at Profile; ?? supplies "Guest."

A	Confusion set	That happens with . but not with ?.
B	Correct	The chain returns null, and ?? replaces it.
C	Dropped element	Ignores the ?? fallback.
D	Adjacent category	The fallback is "Guest," not an empty value.

8. Answer A Concept 6

A product-search endpoint for a hardware retailer, built during a two-week sprint, adds the user's search text directly into a SQL string. Only logged-in users can search. What is the best defensive fix?

Target word: defensive input handling **Decides:** Untrusted text is being mixed into query code.

A	Correct	Parameters keep data separate from query code; allow-lists limit what is accepted.
B	Absolute language	Authenticated input is still untrusted.
C	Adjacent category	Hand-escaping is fragile and easy to get wrong.
D	Dropped element	Block-lists miss variants; data and code are still mixed.

9. Answer C Concept 6 · transfer

An exception is thrown inside a method that checks whether a user may view a record. Which failure behavior is safe?

Target word: safe failure in a security check **Decides:** Errors in authorization must not grant access.

A	Confusion set	Unbounded retries hang the request instead of failing safely.
B	Adjacent category	Leaks internal details to the client.
C	Correct	Fails closed and avoids leaking internals.
D	Inverted operation	Fails open, granting access on error.

10. Answer A Concept 7

A SummaryService class at a legal-tech startup ~~with an award-winning logo~~ creates new OpenAiClient() inside its constructor. Unit tests are slow and cost money on every run. ~~The startup has 14 employees.~~ What design change helps most?

Target word: testable boundary **Decides:** A concrete external client is hard-wired in.

A	Correct	Tests can pass a fake; production passes the real client.
B	Confusion set	Couples the class even more tightly to the concrete client.
C	Adjacent category	Still a hard dependency on the live service.
D	Absolute language	Interface calls add negligible cost.

11. Answer D Concept 8

A checkout bug at a pet-supply store ~~with a popular blog~~ applied a discount twice. A developer fixed it ~~on a Friday afternoon~~. Code coverage is already 85%. What should be added so the bug cannot quietly return?

Target word: regression protection **Decides:** A fixed bug needs an automated test that reproduces it.

A	Dropped element	Not repeated on future changes.
B	Adjacent category	Slow and brittle; a focused unit test pins the logic.
C	Absolute language	Coverage measures lines run, not behaviors checked.
D	Correct	Fails if the bug is reintroduced.

12. Answer A Concept 8 · transfer

In a healthy automated test suite, which type of test should be the most numerous?

Target word: test pyramid shape **Decides:** Fast, isolated tests form the base.

A	Correct	Fast and cheap; they form the base of the pyramid.
B	Confusion set	Useful, but not automated regression checks.
C	Inverted operation	These belong at the narrow top.
D	Absolute language	Cost and speed differ, so counts should differ.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	6	
Wrong stem number	A figure copied from the prompt that is not the answer.	1	
Adjacent category	The right idea in the wrong bucket.	11	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	3	
Absolute language	Always, never, guarantee — where judgment applies.	5	
Confusion set	Every option is real; the test is exact role-matching.	10	
Camouflage	Answers a stem fact that was never the question.	0	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Concurrent awaits with Task.WhenAll	Start independent tasks first and await Task.WhenAll; total time $\approx$ the slowest task, not the sum.	Believing sequential awaits run the calls in parallel.	Several independent I/O calls with given durations.
2	async Task vs. async void	Return Task from async methods so callers can await and catch exceptions; reserve async void for event handlers.	Wrapping a call to an async void method in try/catch and expecting it to catch the error.	An async void method whose exceptions crash the process.
3	Deterministic cleanup with using	A using statement or declaration calls Dispose when scope ends, even if an exception is thrown.	Relying on the garbage collector or a Dispose call at the end of the method.	Files, streams, or connections left open after errors.
4	LINQ deferred execution	Most LINQ queries run when enumerated, not when defined; ToList() or ToArray() takes a snapshot.	Assuming the query result is fixed at the line where it is written.	The source collection changes between defining and using a query.
5	Null-safe operators	?. returns null instead of throwing when the left side is null; ?? supplies a fallback for null.	Expecting a NullReferenceException from a ?. chain, or forgetting ?? supplies the default.	An expression chain with ?. and ??.
6	Secure input handling and safe failure	Treat all input as untrusted: use parameterized queries and allow-list validation, and fail closed with generic error messages while logging details internally.	Trusting authenticated users' input or hand-escaping characters; failing open on errors.	User text reaching a query, or an exception inside a security check.
7	Interfaces and dependency injection	Depend on an interface passed in through the constructor so implementations can be swapped and mocked in tests.	Newing up concrete service clients inside business logic.	A class that is hard to unit test because it calls a live external service.
8	Structured testing strategy	Most tests should be fast unit tests, fewer integration tests, and fewest end-to-end; every fixed bug gets an automated regression test run in CI.	Believing a manual check or high coverage alone prevents regressions.	A bug fix, a coverage number, or a question about which tests to write.

Course scope (public description): This course is the second in a series of three C# courses in BSAIE (AI Engineering with C#>Advanced C#>C# .NET Backend Development). This course builds on prior exposure to programming logic and introductory C# applications, learners develop proficiency with advanced C# language features and core .NET libraries to build reliable, scalable, and maintainable software components. Learners apply asynchronous programming patterns, modern C# constructs, and standard libraries to design production ready components with clear boundaries and contracts that support integration scenarios such as packaging, serving, and consuming external services. The course emphasizes professional software engineering practices, including secure input handling, abuse resistant design, safe failure behaviors, and the implementation of a structured testing strategy with automated validation and regression considerations to support production readiness. These practices prepare learners for backend service design and deployment in the subsequent course.